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16 Types of RAG

Slide to Know More

Standard RAG

Key Feature :

- *Basic retrieval + generation integration*
- *RAG-Sequence and RAG-Token variants*

Benefits :

- *Improved accuracy*
- *Reduced hallucinations*

Applications / Requirements :

- *General-purpose QA systems*
- *Initial RAG implementations*

Examples of Tooling / Library:

- *Hugging Face Transformers*
- *Facebook's RAG implementation*
- *LangChain*

Agentic RAG

Key Feature :

- *Autonomous agents*
- *Tool use*
- *Dynamic retrieval*

Benefits :

- *Handles complex tasks*
- *Proactive AI*

Applications / Requirements :

- *Personal assistants*
- *Research aids*
- *Customer service bots needing dynamic interaction*

Examples of Tooling / Library:

- *LangChain Agents*
- *OpenAI GPT-4 with Plugins*
- *Microsoft Semantic Kernel*

Graph RAG

Key Feature :

- *Knowledge graphs*
- *Relational reasoning*

Benefits :

- *Rich information*
- *Strong context handling*

Applications / Requirements :

- *Expert systems (medicine, law, engineering)*
- *Semantic search engines*

Examples of Tooling / Library:

- *Neo4j Graph Database*
- *Apache Jena*
- *Stardog*

Modular RAG

Key Feature :

- *Independent modules for retrieval, reasoning, generation*

Benefits :

- *Flexibility*
- *Scalability*

Applications / Requirements :

- *Large collaborative projects*
- *Systems needing frequent updates*

Examples of Tooling / Library:

- *Microservices Architecture*
- *Docker & Kubernetes*
- *Apache Kafka*

Memory Augmented RAG

Key Feature :

- *External memory storage & retrieval*

Benefits :

- *Continuity*
- *Personalization*

Applications / Requirements :

- *Chatbots maintaining long-term context*
- *Personalized recommendations*

Examples of Tooling / Library:

- *Redis for session storage*
- *Amazon DynamoDB*
- *Pinecone Vector Database*

Multi Modal RAG

Key Feature :

- *Cross-modal retrieval (text, image, audio)*

Benefits :

- *Richer responses*
- *Accessibility*

Applications / Requirements :

- *Image captioning*
- *Video summarization*
- *Multimodal assistants*

Examples of Tooling / Library:

- *OpenAI CLIP*
- *TensorFlow Hub Models*
- *PyTorch Multi-Modal Libraries*

Federated RAG

Key Feature :

- *Decentralized data sources*
- *Privacy-preserving learning*

Benefits :

- *Data security*
- *Compliance*

Applications / Requirements :

- *Healthcare systems handling sensitive data*
- *Cross-organization collaboration*

Examples of Tooling / Library:

- *TensorFlow Federated*
- *PySyft by OpenMined*
- *Federated Learning Libraries*

Streaming RAG

Key Feature :

- *Real-time data retrieval & generation*

Benefits :

- *Up-to-date information*
- *Low latency*

Applications / Requirements :

- *Live reporting*
- *Financial tickers*
- *Social media monitoring*

Examples of Tooling / Library:

- *Examples of Tooling / Library*
- *Apache Kafka Streams*
- *Amazon Kinesis*
- *Spark Streaming*

ODQA RAG

Key Feature :

- *Broad knowledge base*
- *Dynamic retrieval*

Benefits :

- *Broad applicability*
- *Dynamic responses*

Applications / Requirements :

- *Search engines*
- *Virtual assistants handling diverse queries*

Examples of Tooling / Library:

- *Elasticsearch*
- *Haystack by deepset*
- *Hugging Face Transformers*

Contextual Retrieval RAG

Key Feature :

- *Context-aware retrieval using conversation history*

Benefits :

- *Personalization*
- *Coherence*

Applications / Requirements :

- *Conversational AI*
- *Customer support chatbots maintaining session context*

Examples of Tooling / Library:

- *Dialogflow by Google*
- *Rasa Open Source*
- *Microsoft Bot Framework*

Knowledge Enhanced RAG

Key Feature :

- *Integration of structured knowledge bases*

Benefits :

- *Factual accuracy*
- *Domain expertise*

Applications / Requirements :

- *Educational tools*
- *Professional domain apps (legal, medical)*

Examples of Tooling / Library:

- *Knowledge Graph Embedding Libraries*
- *OWL APIs*
- *Apache Jena*

Domain Specific RAG

Key Feature :

- *Customization for specific industries*

Benefits :

- *Relevance*
- *Compliance*
- *Trustworthiness*

Applications / Requirements :

- *Legal research assistants*
- *Medical diagnosis support*
- *Financial analysis tools*

Examples of Tooling / Library:

- *LexPredict Contract Analytics*
- *Watson Health*
- *Financial NLP Tools*

Hybrid RAG

Key Feature :

- *Combines multiple retrieval approaches*

Benefits :

- *Improved recall*
- *Enhanced relevance*

Applications / Requirements :

- *Complex QA systems*
- *Search engines needing lexical + semantic matching*

Examples of Tooling / Library:

- *Elasticsearch with kNN plugin*
- *FAISS by Facebook AI*
- *Hybrid Retrieval Libraries*

HyDE RAG

Key Feature :

- *Hypothetical document embeddings for guided retrieval*

Benefits :

- *Better recall*
- *Higher answer quality*

Applications / Requirements :

- *Complex queries with implicit meaning*
- *Research assistance in niche fields*

Examples of Tooling / Library:

- *Custom Transformer implementations*
- *Haystack Pipelines*

Recursive Multistep RAG

Key Feature :

- *Multiple rounds of retrieval and generation*

Benefits :

- *Enhanced reasoning*
- *Deeper understanding*

Applications / Requirements :

- *Analytical & problem-solving tasks*
- *Multi-turn dialogue systems*

Examples of Tooling / Library:

- *LangChain chains & agents*
- *DeepMind AlphaCode framework*

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